Hardik Pajiala

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Summary

Computer Engineering undergraduate (2028) in data engineering, applied ML and AI agents. First place at the BIOS 2026 hackathon; built a 2M+ record Delta Lake platform on PySpark and a rule-guarded LLM agent.

Education

Thapar Institute of Engineering & Technology CGPA: 8.50 / 10
Bachelor of Engineering in Computer Engineering *2024 – 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming, Software Engineering, Probability & Statistics, Machine Learning, Deep Learning.

Experience

SafeTrail – First Place, BIOS 2026 Hackathon Thapar Institute, Patiala
Top contributor in a 4-person team | FastAPI, React, TypeScript, MapLibre, LLM APIs *Oct 2026 – Nov 2026*

- Built a trail-safety **PWA** for treks: geofenced danger alerts, **offline-queued SOS**, live responder map.
- Kept AI out of the emergency path: alerts, SOS and triage run as plain code with **Ed25519**-signed passes and event log, while LLM features get per-job budgets, standard-text fallbacks and injection flags; 86 tests.
- Benchmarked 4 builds on a 100-trip simulation: **82% fewer tokens per trip** than a naive build.

Smriti – Smart India Hackathon (SIH26003) MDoNER problem statement
Team project | React, TypeScript, Web Audio API, Flutter, SQLite *Sep 2026*

- Built a memory and care companion for elders with dementia in North East India: an elder screen with read-aloud and a Bihu rhythm game, a **caregiver dashboard**, and an offline **Flutter** tablet build.

Projects

Cricket Win-Probability Lakehouse | *PySpark, Delta Lake, Spark MLlib, Airflow* | Live demo Aug 2026 – Sep 2026

- Engineered a bronze/silver/gold **medallion lakehouse** on Delta Lake turning 8,852 real T20 matches of nested JSON into **2.03M ball-by-ball rows**; an orchestrated full rebuild runs in 3 min.
- Implemented exactly-once ingestion with **Spark Structured Streaming** and idempotent **Delta MERGE** upserts (0 lost records, 0 duplicate keys), guarded by 19 **data-quality checks** that halt downstream tasks.
- Trained a **Spark MLlib** win-probability model (**0.845 AUC**, 1.0 pp calibration error on unseen 2025–26 matches) served live via `applyInPandasWithState`, with **Airflow** DAGs and CI.

Indian Sign Language Recognition | *PyTorch, MediaPipe, BiLSTM* | Live demo Oct 2026

- Built a live webcam recognizer for 61 signs with a **BiLSTM with attention** on MediaPipe keypoints; replaced a saturated benchmark with harder protocols (**95.6%** top-1 from 5 examples per word).

FoodBridge – Backend & Matching Engine | *Node.js, TypeScript, PostgreSQL/PostGIS, FastAPI* Jul 2026 – Sep 2026

- Owned the backend of a 3-person food-redistribution platform: **26 REST endpoints** over a 10-table PostgreSQL schema, with JWT auth, role-based guards and state machines enforcing every donation and pickup transition.
- Designed a **FastAPI** matching service scoring donor–recipient pairs on urgency, proximity and capacity, assigning them optimally with the **Hungarian algorithm** over **PostGIS** spatial distances; covered API and matcher with Vitest and pytest tests.

Academic Timetable Generator | *Java, JavaFX, MySQL/JDBC* Oct 2026

- Modelled timetabling as **graph colouring**: clique and Hall's-theorem checks prove infeasibility early, **backtracking** (MRV, forward checking) meets hard constraints, **simulated annealing** the soft ones.

Technical Skills

Languages: Python, SQL, TypeScript/JavaScript, Java, C/C++, R
Big Data: Apache Spark (PySpark, Structured Streaming, Spark SQL), Delta Lake, Kafka, Parquet, Airflow
ML & AI: PyTorch, TensorFlow/Keras, Spark MLlib, MediaPipe, BiLSTMs, LLM agents & guardrails
Databases & Backend: PostgreSQL, PostGIS, MySQL, Node.js, Express, FastAPI, REST APIs, React, PWAs
Tools: Pandas, NumPy, Matplotlib, Streamlit, pytest, Vitest, Git, GitHub Actions (CI/CD), Linux
Concepts: Medallion lakehouse architecture, ETL/ELT, data modelling, data quality, idempotent upserts, orchestration

Leadership & Extracurricular

Batch Group Leader, Thapar University | *faculty liaison for a 300-student batch* 2025 – Present
Volunteer, Saturnalia (annual college fest) | *coordinated stage-event logistics with 25 volunteers* 2025